Big Data and Machine Learning for Applied Economics

Week 06: Opening the Black Box: What the Machine Learned

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Previously...

- **Split, average, and correct:** a tree splits the stations into groups and averages; a forest averages hundreds of trees; boosting adds small trees, each correcting the previous one.
- **Week 5's answer:** with the best linear regression's same ingredients (census and department), boosting left 12.8 against 15.6. With single-split trees, no interactions, cross-validation gave 15.5; with six, 13.2: the interactions the machine found are worth 2.3 points.
- **And the map:** with longitude and latitude as two more variables, forest and boosting reached 12.1, the course's best so far.

What was left pending

Boosting predicts better than everything we wrote by hand, but it does not say what it learned: which variables it uses, with what shape, which interactions it found, and why it predicts what it predicts at each station.

Today's question: what did the machine learn?

No new model enters today: we enter week 5's

- **The model:** the race-winning boosting (census + department): 738 trees, 38,330 leaves, and a test error of 12.8. In the practical we ran the census-only one, which left 15.7. Nobody reads 738 trees.
- **Today's story, in two questions:** which variables does the model use? (importance: gain, permutation, and ablation) and why does it predict what it predicts at this station? (Shapley and SHAP). In each: the intuition, the math, the algorithm, and the figure.
- **And the control question:** what do these tools *not* tell us? (The shape of the effect, the map, and the error by subgroup live in the appendix, A8–A15.)

The takeaway

Each of today's tools moves one variable and watches what the prediction does. It asks the model, not the world: it says how it predicts, not what would happen if the world changed.

1. Which variables does it use? Importance
2. Why this station? Shapley and SHAP
3. How economists use it: the course-case paper
4. Problem Set 2
5. Closing

The model we are about to open

Week 5's boosting: the 37 census variables and the department, in seven blocks

Block	Columns	What it says about the station's surroundings
Education	3	no education, technical, and higher (% of people)
Age and gender	7	five age groups, women, and voting-age
Ethnicity	5	Indigenous, Rom, Raizal, Palenquero, and Afro
Utilities	5	water, sewerage, power, gas, and internet (% of housing units)
Stratum	7	housing units in each stratum, 0 through 6
Scale and rurality	10	population, housing, blocks, households, distance, registered voters, zone
Department	33	one column of zeros and ones per department

- **Three quantities, all session long:** $\hat{f}(x)$, a station's prediction; $\mathbb{E}[\hat{f}(X)]$, the average prediction or *base* (52.5); and the error **on the test set**: interpreting a model on the data it memorized is interpreting its memory.
- **Notation:** x_{-j} is every variable except j. Nearly everything today is "move x_j and watch what $\hat{f}$ does".
- **The first chapter:** which variables does it use? Three ways to ask...

How much is a variable worth? Three ways to ask

This part's guide: Molnar (2025), *Interpretable Machine Learning*

1. **Gain: what the model says about itself.** Every split of every tree lowered the training loss; x_j's gain adds up what its splits lowered. Free (`xgb.importance`), but measured where the model memorized.
2. **Permutation: damage its input.** Shuffle column x_j on the test set and measure how much the error rises. The model is not touched: the information is taken away without warning it.
3. **Ablation: remove it and retrain.** Train another model without x_j and compare. It measures how much is lost without that information, when the model can defend itself with what is left.

Today's star

Permutation: it is earned on the test set and works for any model. Gain comes along for comparison; ablation returns in the practical.

Importance, formally

Three measures on the same model $\hat{f}$; the error always on the test set, except for gain

Let $e = \mathrm{RMSE}\big(y, \hat{f}(X)\big)$ be the full model's test error (12.8).

- **Gain** (on the training set): $G_j = \sum_{\text{splits on } x_j} \Delta\mathrm{RSS}$, reported as % of the total.
- **Permutation** (on the test set): with $X^{(j)}$ the test set with column j shuffled,

$$\mathrm{VI}_j \;=\; \mathrm{RMSE}\big(y, \hat{f}(X^{(j)})\big) \;-\; e.$$

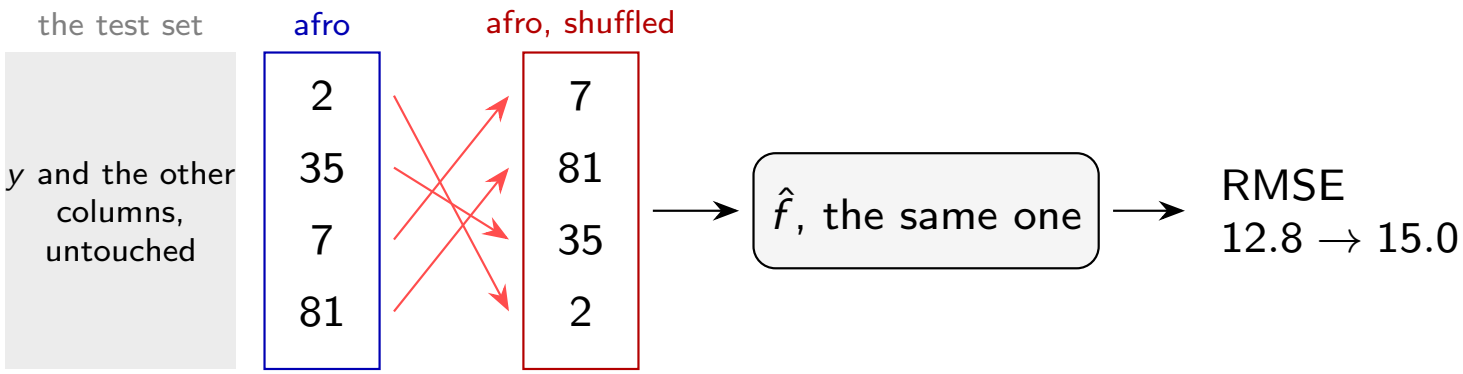
- **Ablation** (retraining): with $\hat{f}_{-j}$ the model trained without x_j,

$$A_j \;=\; \mathrm{RMSE}\big(y, \hat{f}_{-j}(X_{-j})\big) \;-\; e.$$

Three different questions: G_j, what it used to memorize; VI_j, how much *this* model needs x_j to predict; A_j, how much that information is worth when another model can replace it. The ratio is used instead of the difference as well (Fisher, Rudin, and Dominici, 2019).

The permutation algorithm, step by step

Breiman (2001); Fisher, Rudin, and Dominici (2019). The model is not touched: one of its columns gets damaged

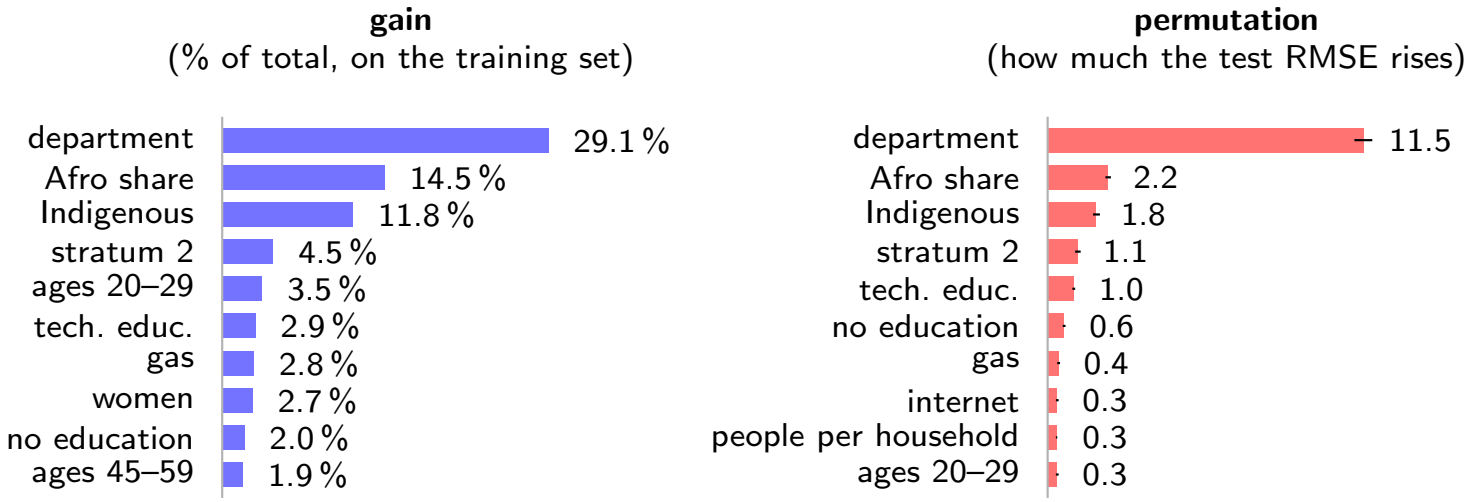


1. **Estimate the best model** (week 5's) and measure its test error: $e = 12.8$.
2. **Shuffle** column x_j: the same values, in another order; its link with y breaks. The model is not refit.
3. **Predict again** and subtract: how much the error rose is VI_j.
4. **Repeat** $M = 10$ times and average: one shuffle is noisy.
5. **Repeat for each** $j = 1, \ldots, p$ and sort: the *ranking* on the next slide.

Our own detail: the department is shuffled whole (its 33 columns together, same row order): shuffling them separately would fabricate polling stations with two departments.

The two importances, side by side

Gain looks at the training set; permutation, at the test set



- **How to read it:** on the left, gain (% of the total, measured on the training set); on the right, permutation (how much the test RMSE rises, in points). Black lines: one standard deviation across the ten shuffles.
- **The verdict:** shuffling the department raises the error 11.5 points; the Afro share, 2.2; the Indigenous share, 1.8. Higher education, week 2's star, barely 0.25.
- **They agree at the top and part ways below:** memorizing the training set is not predicting the test set. Before trusting any *ranking*: three traps.

Importance's traps

1. **Permuting forces extrapolation** (Hooker, Mentch, and Zhou, 2021). Shuffling the Afro share while holding the department fixed manufactures Chocó stations with no Afro population, which do not exist.
2. **Credit gets split.** Two variables telling the same story look small on their own and large together: they take turns at the splits and each takes half.
3. **It depends on the model.** Two models with the same error can lean on different variables (appendix A13).
4. **It is unstable.** It changes with the seed, the resample, and the shuffles: report it with bars.

And the usual one

Important is not causal. The most important variable may have no effect at all: it only needs to be the best proxy for what does.

Trap 2 has a treatment: measure whole blocks (appendix A12) and ablation, which we run today in the practical.

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Is it Chocó or is it the Afro-descendant share?

A station in Quibdó, three players, and one regression per coalition

Colegio Pedro Grau Arola (Quibdó; voted 88.8 for Petro). The game: $v(S)$ is the prediction of a regression with the blocks in S, minus the average (52.5). The players: **E**ducation, e**T**hnicity, and the **D**epartment.

S	$\emptyset$	E	T	D	ET	ED	TD	ETD
$v(S)$	0.0	-1.1	31.2	33.0	33.2	34.7	32.2	34.6

Order of entry	E adds	T adds	D adds
E, T, D	-1.1	34.3	1.4
E, D, T	-1.1	-0.1	35.8
T, E, D	2.1	31.2	1.4
T, D, E	2.4	31.2	1.1
D, E, T	1.7	-0.1	33.0
D, T, E	2.4	-0.8	33.0
Average, ϕ	**1.1**	**15.9**	**17.6**

Shapley's idea (1953): let the players enter in some order, write down how much each adds on entry, and average over all orders.

Question for the room: how much of this station's vote is Chocó and how much is the Afro-descendant share?

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It depends on the order: whoever enters first takes almost everything (31.2 or 33.0 points) and the second, almost nothing, because they tell the same story. Shapley splits it: 15.9 to ethnicity, 17.6 to the department, and 1.1 to education; they add to 34.6, the distance between the prediction (87.1) and the average.

The Shapley value: the formula and why it is unique

$$\phi_j \;=\; \sum_{S \subseteq \{1,\dots,p\}\setminus\{j\}} \frac{|S|!\,(p-|S|-1)!}{p!}\Big[v(S\cup\{j\})-v(S)\Big]$$

The weight is the fraction of orders in which j enters right after those in S: the previous table, written in general.

It is the only splitting rule with four reasonable properties (appendix A1):

- **Efficiency:** $\sum_j \phi_j = v(\text{all}) - v(\emptyset)$. All the credit is handed out.
- **Symmetry:** two interchangeable players receive the same.
- **Null player:** whoever never adds anything receives zero.
- **Additivity:** the split of the sum of two games is the sum of the splits.

From blocks to columns

With p columns there are 2^p coalitions: with our model's 70, a 22-digit number. And you must decide what "using only those in S" means without retraining 2^p models.

SHAP: Shapley for machine learning models

Lundberg and Lee (2017); Lundberg et al. (2020); the guide: Molnar (2025), ch. SHAP

The local explanation is additive: for a station x,

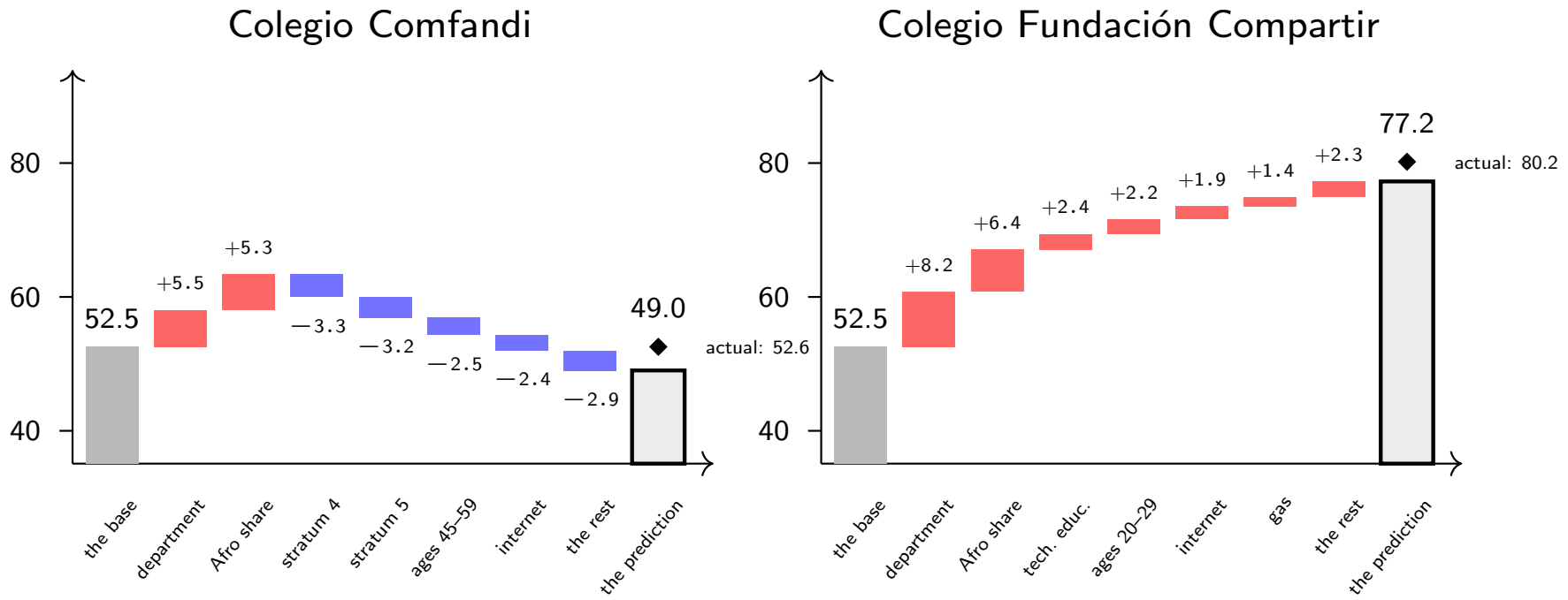
$$\hat{f}(x) = \underbrace{\mathbb{E}[\hat{f}(X)]}_{\phi_0,\ \text{the base}} + \sum_{j=1}^{p} \phi_j(x),$$

and each $\phi_j(x)$ is column j's Shapley value in the game "predict x".

- **What $v(S)$ is for a model:** the expected prediction when only the columns in S are known; the absent ones are filled with background data, ignoring the correlation (*marginal*) or honoring it (*conditional*). Appendix A2.
- **How it is computed:** KernelSHAP samples coalitions (any model; slow; appendix A3). **TreeSHAP** is exact and fast on trees: the 2,398 test stations in seconds (`predcontrib = TRUE` in `xgboost`).
- **Verified station by station:** base (52.5) + sum of contributions = prediction, with a maximum error of 0.0002 points (machine rounding).

Two schools in Cali, 28 points apart

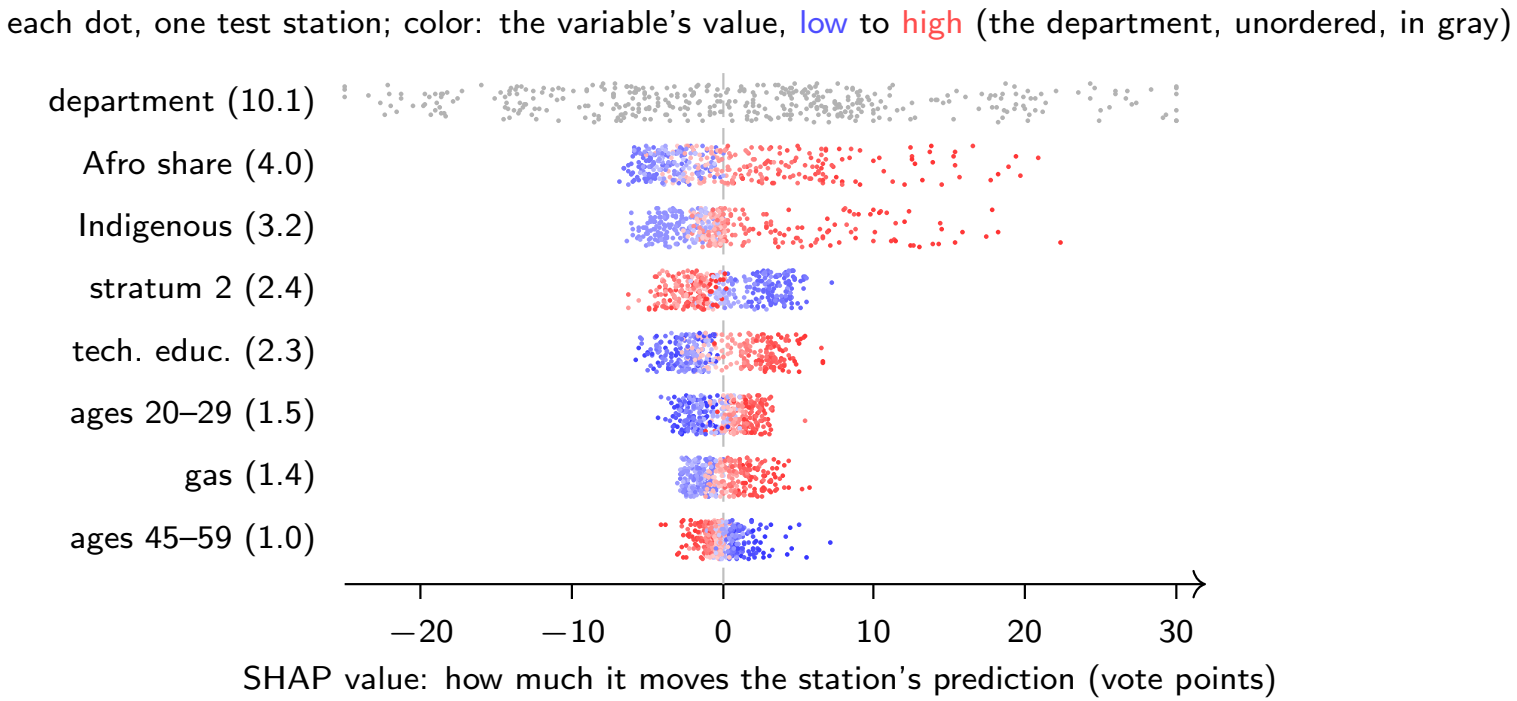
Each prediction's SHAP split, from the base to the prediction: the six columns that move it most



- **Colegio Comfandi:** 41 % higher education and 78 % of housing in strata 4 and 5, which pull it down. **Colegio Fundación Compartir:** 71 % in stratum 1 and a 35 % Afro share.
- **The same department weighs differently** (5.5 and 8.2): with no interactions it would weigh the same in both. It is an interaction, seen at one station.

From local to global: every station at once

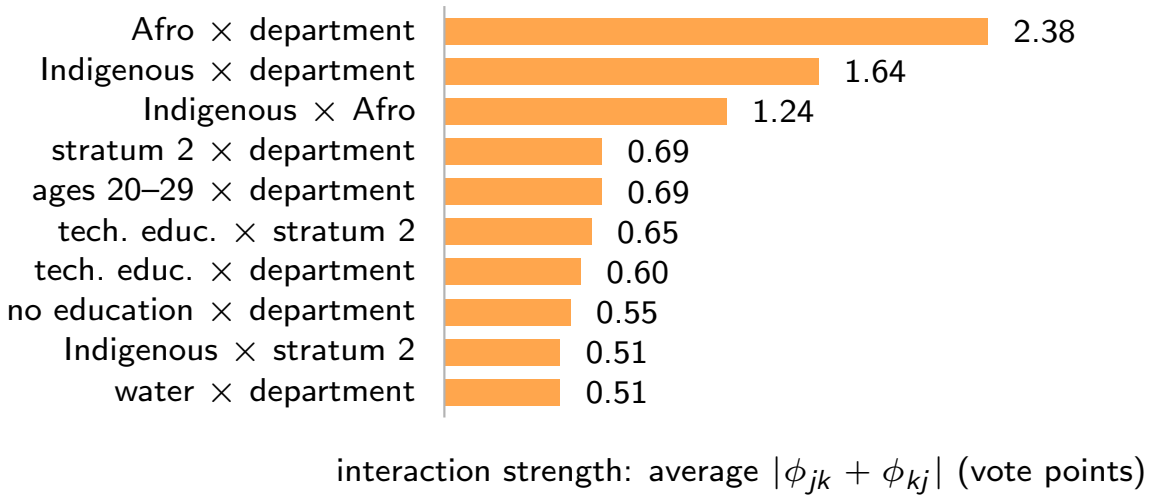
The beeswarm: one row per variable, one dot per test station; in parentheses, the average |SHAP|



- **Global importance** is the average absolute value: department 10.1, Afro share 4.0, Indigenous 3.2. The same hierarchy as permutation, with a local reading behind it.
- **Color gives the direction** and **the spread within a row is the interaction:** the same variable value pushes differently depending on the rest of the station.

The interactions the machine found

SHAP interaction values at 500 test stations: the credit that only exists when two variables move together



- A station's SHAP split divides into each variable's own effect and one per pair, ϕ_{jk}: what the model does with x_j *because* it saw x_k (Lundberg et al., 2020; appendix A6).
- **The answer to week 5's pending question:** the interactions worth 2.3 points are, above all, ethnicity within each department: exactly the slopes week 3 refused to write by hand and the machine found here on its own.

What a SHAP value is not

1. **It is not a causal effect.** It splits a prediction across the model's columns: the Afro share does not "cause" its points; it is the column that best summarizes what the model learned about the Pacific.
2. **It is not unique.** It changes with the fill-in, the background, and the model (appendix A13): report what it was computed with.
3. **It is not about the voter.** The station is an aggregate: ecological inference (Robinson, 1950).
4. **It is not a forecast.** It explains the 2022 election; that it repeats in 2026 is a hypothesis.
5. **It does not replace the test set.** A pretty explanation of a bad model is a story about noise.

Rudin (2019), in one sentence

Explaining a black box *afterwards* does not make it transparent: when the decision demands it, compare with a model interpretable by design and measure how much is lost.

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How economists use it: the course-case paper

Gelvez, Cardiles, Martínez-González, and Muñoz (2026): how predictable is an election?

- **The question:** how much of an election can be predicted from the social and territorial structure of places? The paper calls it *structural predictability*, and measures it the way we do here: with models evaluated on data they never saw.
- **The data is ours:** Colombia's polling stations, 2006–2022, with the census around each one; four outcomes per election and round: did the national winner carry this station?, the winner's share, turnout, and the between-round vote gain.
- **The method is the course's:** random forests against regression (logistic or linear), a held-out test set, five-fold cross-validation for the knobs, and SHAP (TreeExplainer) to open the model.

Why we read it today

It is the course case, told in reverse: we use their data; they answer this session's question with SHAP, at the scale of five elections.

The course-case paper: the main results

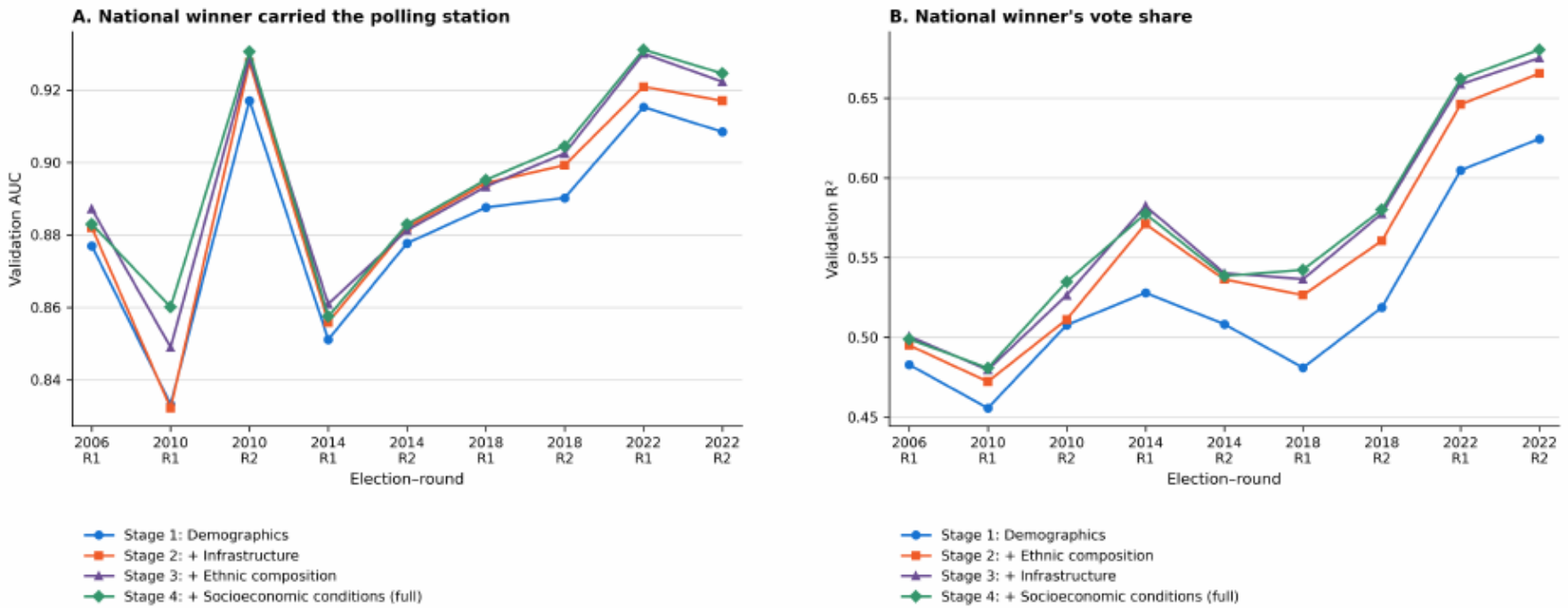
Their Table 1: flexible versus linear, same predictors, same test stations

		Winner here? (AUC)		Winner's share (R^2)	
Election	Round	Logit	Forest	Linear	Forest
2006	1	0.855	0.883	0.385	0.499
2010	1	0.825	0.860	0.374	0.481
2010	2	0.903	0.931	0.376	0.535
2014	1	0.844	0.858	0.493	0.577
2014	2	0.858	0.883	0.469	0.538
2018	1	0.867	0.895	0.442	0.542
2018	2	0.867	0.904	0.439	0.580
2022	1	0.909	0.931	0.578	0.662
2022	**2**	**0.905**	**0.925**	**0.605**	**0.680**

- **How to read it:** one row per election and round, everything evaluated on held-out stations; the bold row is our class's election.
- **Two course lessons:** information does most of the work (the logit already posts 0.905 in 2022) and the flexibility premium is a measurement, not an assumption. And 2022 is the most predictable election in the series.

The course-case paper: predictability and its blocks (their Figure 5)

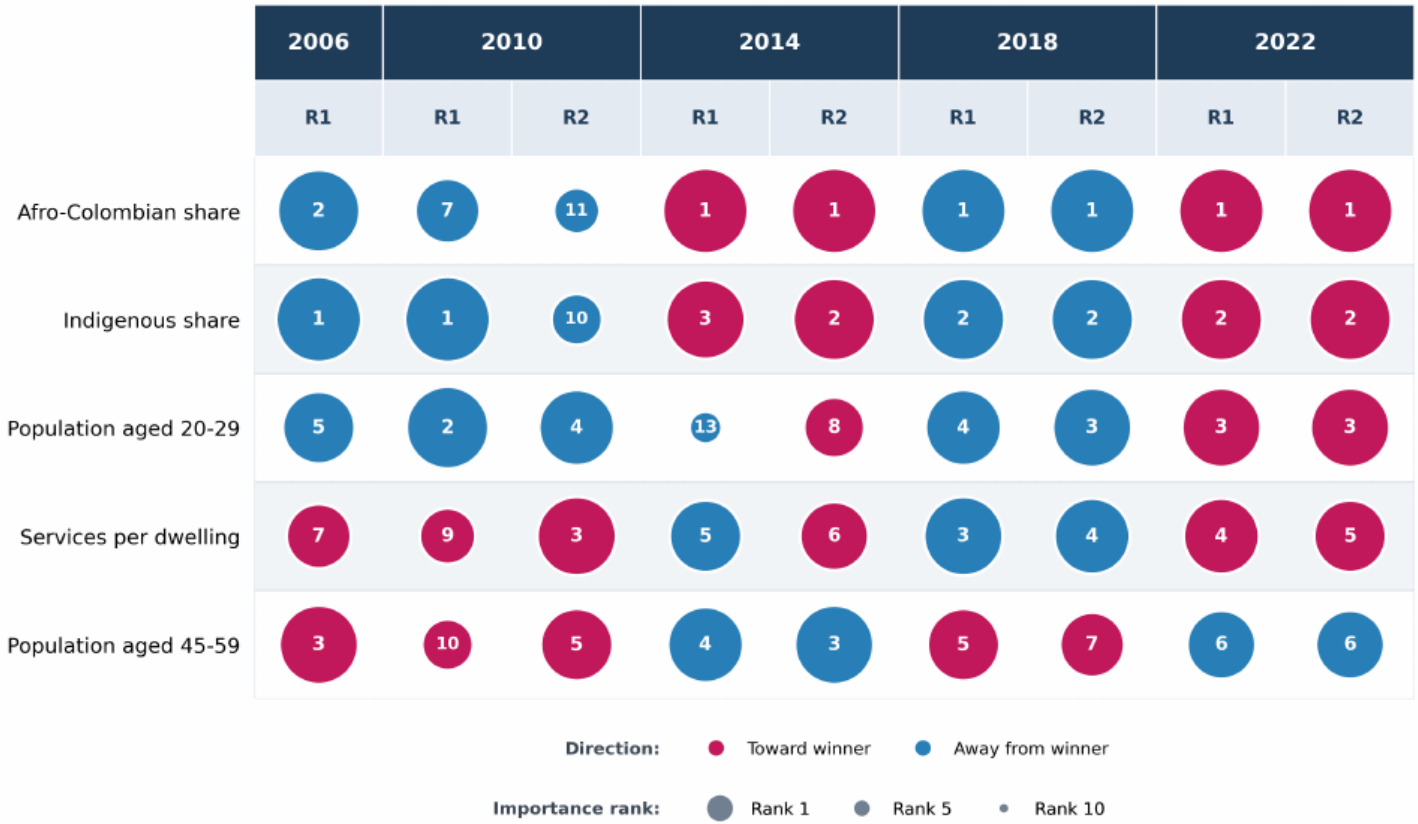
Out-of-sample performance by election, as the forward-selection blocks enter



How to read it: panel A is the AUC for winner-here?; panel B, the R^2 for the winner's share, both on held-out stations. Each line adds one block in the selection order (department dummies always included): the blue line, demographics alone, already does most of it, and 2022 is the peak of the series.

The paper's SHAP (I): the ranking and the direction (their Figure 6)

Within each model: the SHAP importance rank (the size) and which way it pushes (the color)



How to read it: red pushes toward the winner; blue, away. The Afro-descendant share moves to rank 1 in every round from 2014 on, with the Indigenous share at 2–3: the same two that lead our 2022 model.

The paper's SHAP (II): the sign changes owners

The same territorial structure, different coalitions: why "important is not causal"

- **The twist:** higher Afro and Indigenous shares push predictions *away* from the winner in 2006, 2010, and 2018 (the Uribe and Duque coalitions) and *toward* it in 2014 (Santos) and 2022 (Petro).
- **The clean contrast:** Santos wins in 2010 and in 2014; ethnicity is negative for his geography in 2010 and positive in 2014. The coalition was transformed by the peace process; the territorial structure was not.
- **And in 2010-II there is barely a boundary left to explain:** Santos carried 87.8 % of the held-out stations, and the ethnic ranks fall to 11 and 10: SHAP splits the boundary that remains, not the whole country.

The lesson for the kit

SHAP says what the model uses and which way it pushes; who benefits from that divide is decided by coalitions. Stable variables, realigning signs: Y, not β.

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Problem Set 2 is released today

From trees to the course's final model, with Cali's housing. Due: Tuesday, November 10, 11:59 p.m.

Same dataset and same rules as Problem Set 1: Cali's listings with their fixed split; the test price stays hidden and opens exactly once, for the course's final model.

Part	Session	Pts	What you do
1. Trees, forests, and boosting	5	20	CV-pruned tree; forest (*mtry*, OOB vs. validation); boosting with early stopping; table against PS1's best model
2. Opening the box	6	25	permutation vs. what the Lasso kept; PDP and ICE; global SHAP and two listings'; block ablation
3. The network against the forest	7	10	a small perceptron, early stopping, three seeds: does it beat the forest?
4. Final model and reading	5–7	15	the best model of both problem sets $\rightarrow$ test once; one page: what the model learned about the housing market and what it does not say
Course ranking		20	by the final model's test RMSE; first place 20, last 10
Reproducibility		10	`00_run.R` runs start to finish

Parts 1 and 2 are covered as of today; part 3, on October 27. **Debrief** of the PS1 ranking in session 7 and of PS2 on November 24.

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What the opened box says and what it does not

- **It says what the prediction depends on:** the department first and ethnicity next; the census blocks matter as a bundle: all used, almost none irreplaceable.
- **It says with what shape** (appendix A8–A10): the Afro share pushes upward; education, nearly flat country-wide and with opposite slopes in Cali and Chocó.
- **It says which interactions it found:** above all ethnicity within each department: the slopes week 3 left unwritten.
- **It says where it fails** (appendix A14): in rural areas, at the extremes, and in departments where the census says little.
- **It does not say why,** nor what would happen if something changed, nor anything about the voter; and it changes with the model (appendix A13).

Y, not β, with the box open

The Afro share's SHAP is not a β: it answers "how does the model use this variable to predict?". Moving to "what would happen if it changed?" demands an identification design: the Causal Inference course.

When to use what

	Use when	Watch out for
Gain or impurity	a free first look	cardinality; measured on the training set; never report it alone
Permutation	ranking variables in any model	extrapolates under correlation; unstable; shuffle whole blocks
Ablation	asking about one type of information	retrains; measures the irreplaceable, not the used
PDP and ICE	the shape of the effect	the average hides heterogeneity; extrapolates
ALE	the shape, with correlated variables	less intuitive; not local
SHAP	coherent local and global explanation	the fill-in; slow without trees; not causal
SHAP interactions	which interactions the model found	expensive: one matrix per station
Error by group	always	leaving it out

What to read: Molnar (2025), with code, for the permutation, PDP, ALE, and SHAP chapters; Lundberg and Lee (2017) for the unification; Lundberg et al. (2020) for TreeSHAP; Rudin (2019) for the objection.

Recap

1. **Interpreting** is asking the model how it predicts, always on data it never saw. Global (the model on average) and local (one station).
2. **Importance:** gain is free and measured where the model memorized; permutation is earned on the test set with the shuffling algorithm (the department: $+11.5$); ablation retrains without the variable (without the department you get week 5's census-only model: $+2.9$).
3. **Shapley:** the only split of a prediction with the four properties; in Quibdó, 15.9 to ethnicity and 17.6 to the department.
4. **SHAP:** TreeSHAP computes it exactly for trees; two schools in Cali where the same department weighs differently, the *beeswarm*, and the interactions: ethnicity within each department.
5. **What they do not say:** they are not causal and they belong to *this* model (Rashomon, appendix A13); the error by subgroup (A14) always goes in the report.

Getting ready for next class...

To get the most out of next session

This week's readings:

- ISL, sections 12.2 (principal components) and 12.4.1 (k-means); sections 10.1 to 10.3 and 10.6 to 10.7 (neural networks).
  Guide: figure 12.1 (the first two components) and 10.4 (the digits network) are the ones from class; 10.6 is the honest section: when to use deep learning.

Problem Set 2: released today; due Tuesday, November 10 at 11:59 p.m. Session 7 opens with the **Problem Set 1 ranking debrief**.

Next session — representing: today we saw that the census blocks stand in for one another. How much distinct information is in the 37 columns? Without a y, PCA compresses them into a few indices and k-means groups the stations into territory types. With a y, a neural network learns its own intermediate variables.

R application: the full repeat of week 5's model (its cross-validation and its 738 rounds); gain; one permutation example and one ablation example (the model without the department is week 5's census-only model); and SHAP with `predcontrib`.

Essential references

- Molnar, C. (2025). *Interpretable Machine Learning* (3. ed.). https://christophm.github.io/interpretable-ml-book/
- Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32.
- Breiman, L. (2001). Statistical Modeling: The Two Cultures. *Statistical Science*, 16(3), 199–231.
- Fisher, A., Rudin, C., & Dominici, F. (2019). All Models Are Wrong, but Many Are Useful. *JMLR*, 20(177), 1–81.
- Friedman, J. (2001). Greedy Function Approximation: A Gradient Boosting Machine. *Annals of Statistics*, 29(5), 1189–1232.
- Goldstein, A., Kapelner, A., Bleich, J., & Pitkin, E. (2015). Peeking Inside the Black Box: ICE. *JCGS*, 24(1), 44–65.
- Apley, D., & Zhu, J. (2020). Visualizing the Effects of Predictor Variables in Black Box Models. *JRSS-B*, 82(4), 1059–1086.
- Lundberg, S., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *NeurIPS*, 30.
- Lundberg, S., et al. (2020). From Local Explanations to Global Understanding with Explainable AI for Trees. *Nature Machine Intelligence*, 2, 56–67.
- Rudin, C. (2019). Stop Explaining Black Box Machine Learning Models for High Stakes Decisions. *Nature Machine Intelligence*, 1, 206–215.
- Gelvez, J. D., Cardiles, A., Martínez-González, E. F., & Muñoz, M. (2026). How Predictable Is an Election? Working paper. [The source of the case data.]

6 Appendix: going deeper

A1. Why the Shapley value is unique

A **game** is a function $v : 2^P \to \mathbb{R}$ with $v(\emptyset) = 0$ on the subsets of $P = \{1, \ldots, p\}$. A **splitting rule** assigns $\phi_j(v)$ to each player.

1. Unanimity games form a basis. For nonempty $T \subseteq P$, let $u_T(S) = \mathbb{I}\{T \subseteq S\}$. Any v is written uniquely as $v = \sum_T c_T\, u_T$: there are $2^p - 1$ games and $2^p - 1$ coefficients.

2. In a unanimity game the split is forced. Players outside T are null ($\phi_j = 0$); those in T are symmetric, and by efficiency they share $u_T(P) = 1$: $\phi_j(u_T) = 1/|T|$ for $j \in T$.

3. Additivity extends to every game: $\phi_j(v) = \sum_{T \ni j} c_T/|T|$. A single rule satisfies the four axioms, and expanding the c_T gives the marginal-contributions formula:

$$\phi_j = \sum_{S \subseteq P \setminus \{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \Big[v(S \cup \{j\}) - v(S) \Big].$$

The Shapley decomposition of R^2 (Shorrocks, 2013): the game is $v(S) = R^2$ of the model with the blocks in S; with seven blocks, 2^7 models. It is the order-free version of part 1's ablation.

A2. Filling in the absent: marginal and conditional SHAP

A prediction's game needs $v(S)$ for every subset of variables. Two definitions:

$$v^{\text{marg}}(S) = \mathbb{E}_{X_{-S}}\left[\hat{f}(x_S, X_{-S})\right], \qquad v^{\text{cond}}(S) = \mathbb{E}\left[\hat{f}(x_S, X_{-S}) \,\middle|\, X_S = x_S\right].$$

- **Marginal:** sample X_{-S} from a background set without looking at x_S. True to the model: a variable $\hat{f}$ does not use receives exactly zero. The price: it evaluates $\hat{f}$ at impossible combinations (appendix A10's stations that do not exist).
- **Conditional:** sample X_{-S} given x_S. True to the data: no extrapolation. The price: a variable the model ignores receives credit if it is correlated with one it does use, and estimating the conditional is hard (Chen, Janizek, Lundberg, and Lee, 2020).
- TreeSHAP has both variants: the "interventional" one, which is the marginal with a background set, and the "path-dependent" one, which approximates the conditional with the nodes' frequencies; `predcontrib` in `xgboost` uses the second. Report which one you used.

A3. KernelSHAP: Shapley as a weighted regression

Lundberg and Lee (2017): the Shapley value is the solution of a weighted linear regression. With $z \in \{0,1\}^p$ indicating which variables are "present" and $v(z)$ the expected prediction with only those,

$$\min_{\phi_0,\phi} \sum_z \pi(z) \left[v(z) - \phi_0 - \sum_j \phi_j z_j \right]^2, \qquad \pi(z) = \frac{p-1}{\binom{p}{|z|}\,|z|\,(p-|z|)},$$

with $|z|$ the number of ones.

- The weight π is huge for nearly empty or nearly full coalitions and small in the middle: it is exactly the one that makes the least squares match Shapley's formula.
- In practice a few hundred or thousand z's (out of 2^p) are sampled and the least squares is solved: approximate, model-agnostic, and slow, because each z requires evaluating $\hat{f}$ over the background set.
- **TreeSHAP** (Lundberg et al., 2020) avoids the sampling: it walks each tree keeping count of how many observations pass through each branch, and gets the exact ϕ_j's in time polynomial in the depth.

A4. PDP, marginal plot, and ALE, in formulas

For a predictor x_j and the rest x_{-j}, three curves answering different questions:

$$\text{PDP:} \quad f_{\text{PD}}(z) = \mathbb{E}_{X_{-j}}\big[\hat{f}(z, X_{-j})\big] \quad \text{(average over the marginal of } X_{-j}\text{)}$$

$$\text{Marginal:} \quad f_{\text{M}}(z) = \mathbb{E}\big[\hat{f}(X) \,\big|\, X_j = z\big] \quad \text{(average over the conditional)}$$

$$\text{ALE:} \quad f_{\text{ALE}}(z) = \int_{x_{\min}}^{z} \mathbb{E}\left[\frac{\partial \hat{f}(X)}{\partial X_j} \,\middle|\, X_j = u\right] du \; - \; \text{const.}$$

- The PDP extrapolates: it averages over values of X_{-j} that never occur with that z. The marginal plot does not extrapolate but it *mixes*: if x_j and x_k move together, x_k's effect sneaks into x_j's curve.
- The ALE averages the **derivative** conditional on $X_j = u$ (no extrapolation) and then **accumulates**; since it differentiates first, x_k's effect disappears. In practice, with K bands of x_j with edges $z_0 < z_1 < \cdots < z_K$:

$$\hat{f}_{\text{ALE}}(z) = \sum_{k=1}^{k(z)} \frac{1}{n_k} \sum_{i:\, x_{ij} \in (z_{k-1}, z_k]} \Big[\hat{f}(z_k, x_{i,-j}) - \hat{f}(z_{k-1}, x_{i,-j})\Big],$$

centered at zero. With independent predictors, the three curves coincide.

A5. Conditional and block permutation, and Rashomon importance

- **Conditional permutation** (Strobl et al., 2008): permute x_j only *within* groups defined by the variables correlated with it. It measures x_j's importance *given* x_{-j}, like a partial coefficient; the unconditional one measures the marginal.
- **Block permutation:** permute a block's columns together (the five ethnicity ones; the department's 33). It answers "how much does the model lose without this type of information?" and avoids the split between twins.
- **Ablation** (*leave-one-covariate-out*): refit without x_j or without the block and compare errors. No extrapolation, but it costs one fit per block and measures something else: the loss when *another* model replaces x_j with its proxies.
- **Model class reliance** (Fisher, Rudin, and Dominici, 2019): the range of x_j's importances across all near-optimal models (the Rashomon set). If it includes zero, no data forces using x_j: "the model needs ethnicity" speaks of this model, not of the problem.
- **Shapley regressions** (Joseph, 2019): regress y on each variable's Shapley values to give them a standard error. They say whether the contribution aligns with the outcome, not whether it causes it.

A6. SHAP interaction values and Friedman's H

SHAP interaction (Lundberg et al., 2020): a pair's Shapley value, with the second difference instead of the first,

$$\phi_{jk} = \sum_{S \subseteq P \setminus \{j,k\}} \frac{|S|!\,(p - |S| - 2)!}{2\,(p-1)!}\,\nabla_{jk}(S),$$

$$\nabla_{jk}(S) = v(S \cup \{j,k\}) - v(S \cup \{j\}) - v(S \cup \{k\}) + v(S).$$

- $\nabla_{jk}(S) = 0$ if j's contribution does not depend on whether k is in: no crossing, no shared credit.
- The split is completed with the own effect, $\phi_j = \phi_{jj} + \sum_{k \neq j} \phi_{jk}$. TreeSHAP computes it exactly (`predinteraction = TRUE`): one $p \times p$ matrix per station, hence today's 500 stations.
- **Friedman and Popescu's H (2008)** is the global version built from PDPs:

$$H^2_{jk} = \frac{\sum_i \left[\hat{f}_{jk}(x_{ij}, x_{ik}) - \hat{f}_j(x_{ij}) - \hat{f}_k(x_{ik})\right]^2}{\sum_i \hat{f}_{jk}(x_{ij}, x_{ik})^2},$$

with the curves centered: zero is additivity; one, pure interaction.

A7. SHAP for gana_petro: the log-odds scale

- Week 5's classification boosting (census and department; test AUC 0.941) predicts on the **log-odds** scale: the base is 0.13 and the SHAP values add up there,

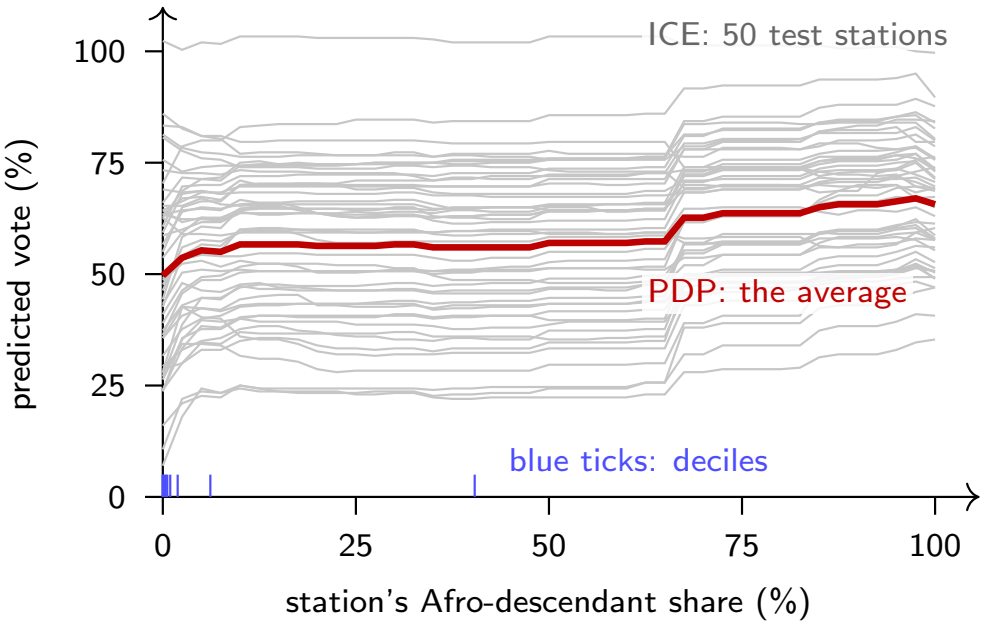
$$\log \frac{\hat{p}(x)}{1-\hat{p}(x)} \;=\; \phi_0 + \sum_{j=1}^{p} \phi_j(x), \qquad \hat{p}(x) = \frac{1}{1+e^{-(\phi_0+\sum_j \phi_j(x))}}.$$

- **What moves the probability that Petro wins the most:** department (1.29 in log-odds, on average), Afro share (0.44), and stratum 2 (0.34): the same hierarchy as in the vote.
- **Careful when translating:** the same ϕ_j moves the probability a lot near 0.5 and almost nothing near 0 or 1. It is week 4's logistic $\beta\, p(1-p)$ effect.

A8. Partial dependence and its individual curves

Friedman (2001); Goldstein et al. (2015). The Afro-descendant share, fixed at each value for the 2,398 test stations

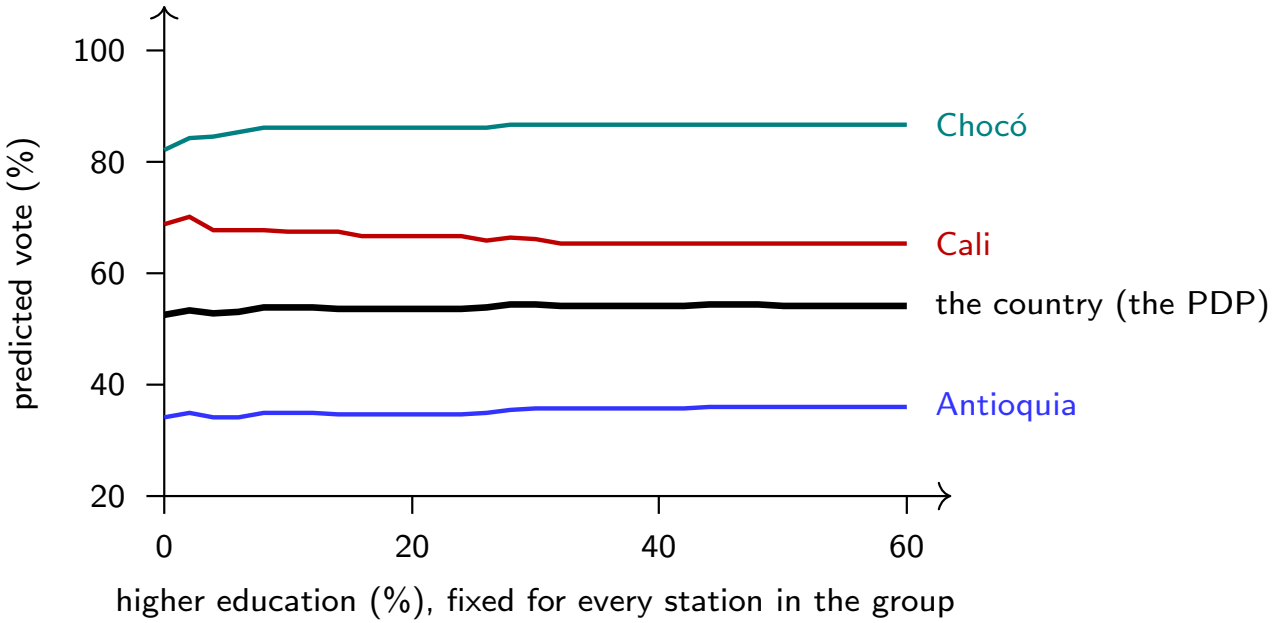
$$\text{PDP:} \quad \hat{f}_j(z) = \frac{1}{n}\sum_{i=1}^{n} \hat{f}\left(z, x_{i,-j}\right) \qquad \text{ICE}_i(z) = \hat{f}\left(z, x_{i,-j}\right)$$



- **PDP:** fix $x_j = z$ at every station, predict, and average: from 49.6 at 0 % Afro share to 65.8 at 100 %.
- **ICE:** the same, one curve per station. Nearly parallel: the Afro share adds the same on top of each station's level.
- **Blue ticks:** the deciles; 80 % of the stations have less than 6.2 % Afro share.

A9. Cali's star, country-wide

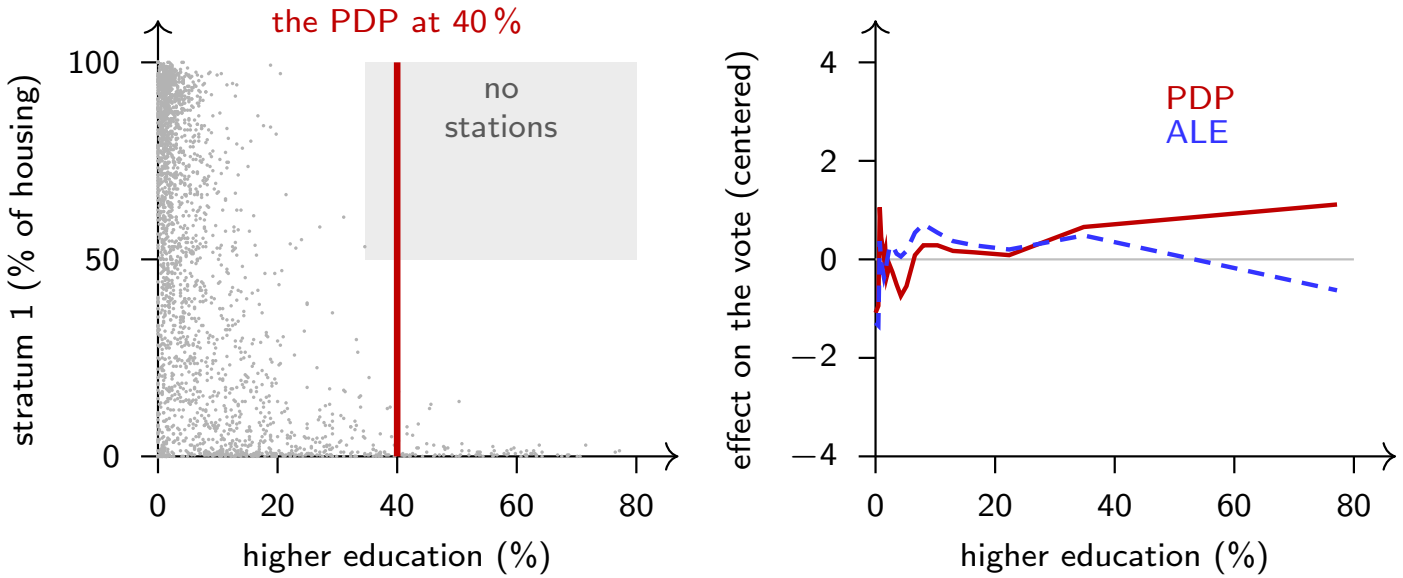
Higher education: the country's curve and the average ICE curves of three groups of stations



- **Country-wide, nearly flat:** the PDP runs from 52.4 to 54.4. Week 2's star variable barely moves the prediction.
- **But the average hides opposite slopes:** at Cali's 43 test stations, raising education *lowers* the prediction (from 68.8 to 65.2); at Chocó's 79, it *raises* it (from 82.1 to 86.6). Slopes that flip sign with place are an interaction: week 3's funnel, found without writing it.

A10. When the PDP asks about stations that do not exist

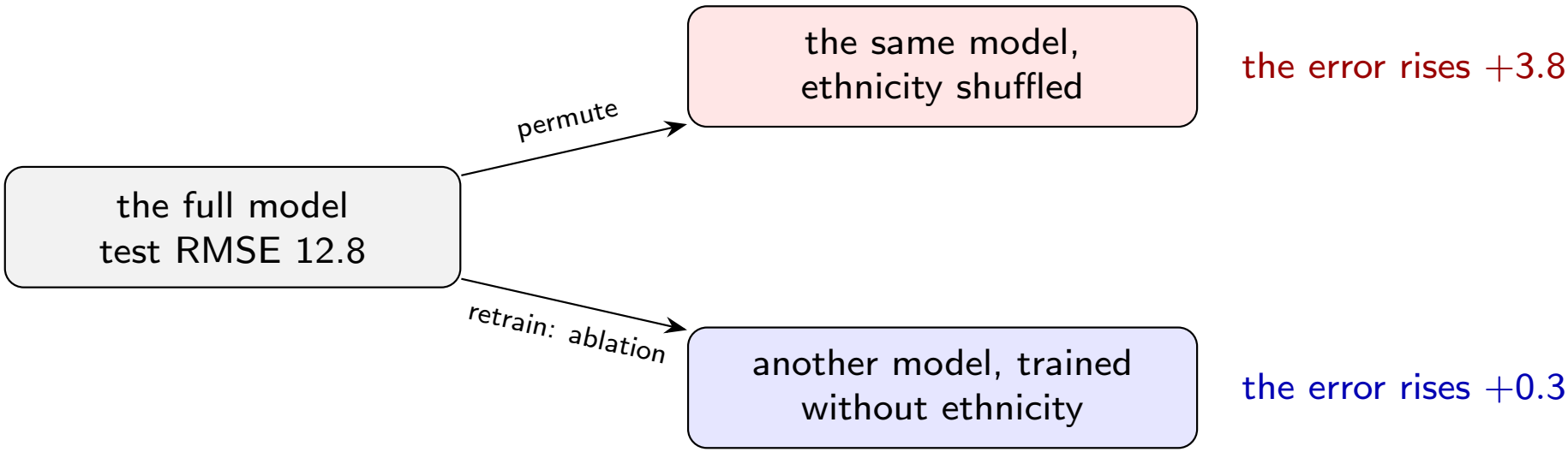
Left: higher education against stratum 1. Right: the education PDP and ALE (Apley and Zhu, 2020)



- **The PDP at 40 %** sets all 2,398 stations at 40 % higher education, including the 1,208 (50 %) with more than half their housing in stratum 1. No such station exceeds 34.7 %: the model predicts for stations that do not exist. It is week 2's extrapolation, hidden inside an average.
- **The ALE** looks only at local differences: within each education band, how much $\hat{f}$ changes between the edges for the stations actually there, then accumulates (appendix A4). This time they agree (maximum difference: 1.7 points); with more correlated variables, the PDP can invent a shape.

A11. Permuting versus retraining: two different questions

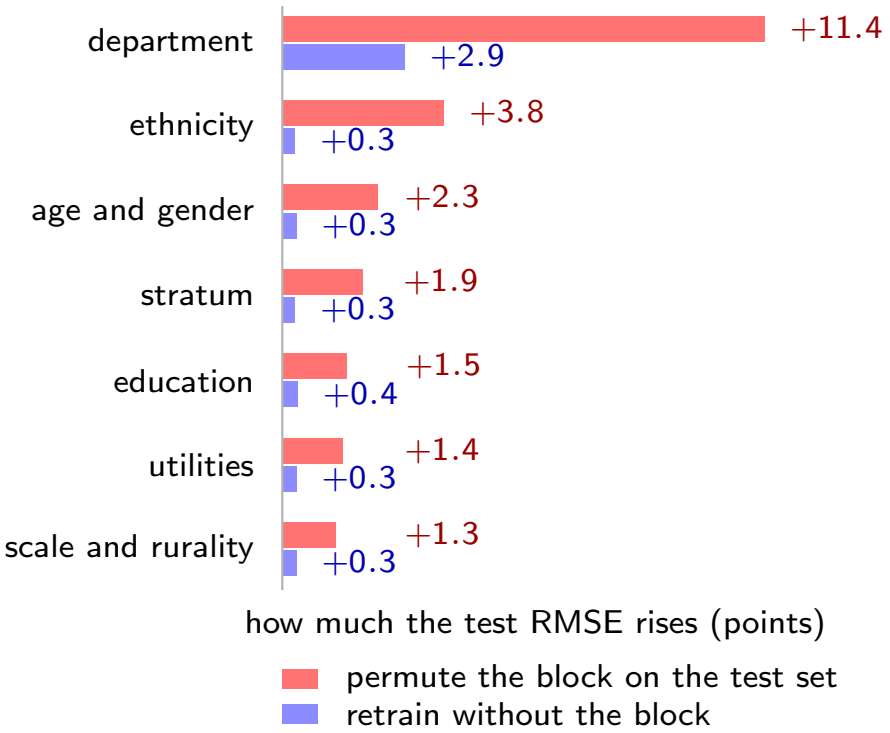
The same block, two operations: damage the model's input, or train another model without it



- **Permuting** answers: how much does *this* model use that block? The model stays intact and its input gets damaged.
- **Retraining without the block** (the *ablation*) answers: how much is lost without that information? A model is trained from scratch, same recipe, and the new one defends itself with what is left.
- **The gap between 3.8 and 0.3 is redundancy:** the other blocks tell the same story as ethnicity, and the retrained model recovers it another way.

A12. The seven blocks, under the two questions

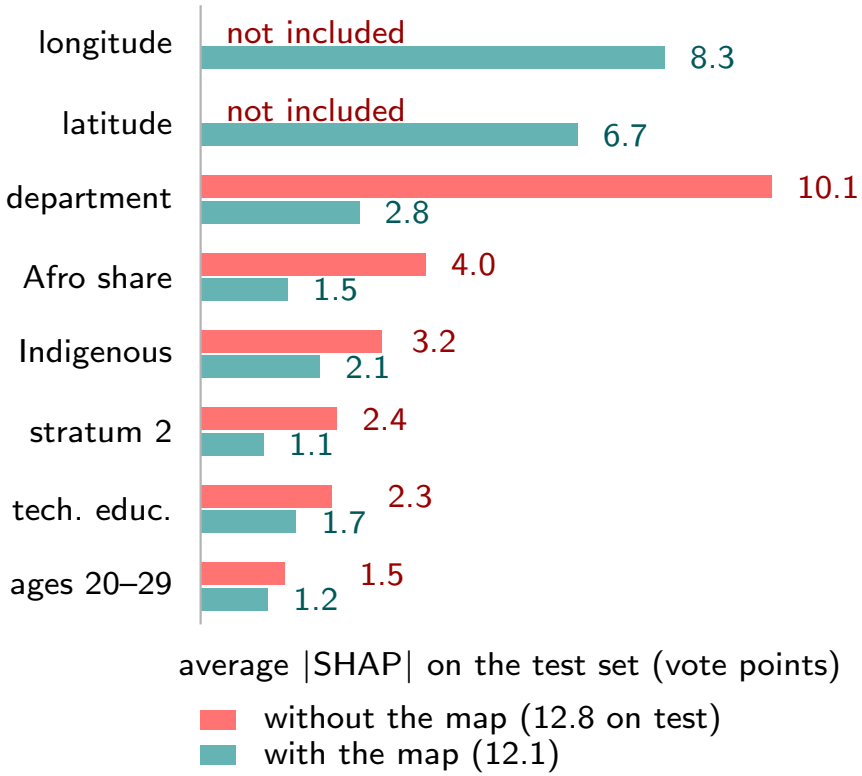
Permutation asks about the model we have; ablation, about the best model without that block



- **How to read it:** each block carries two bars: in red, permuting it on the test set; in blue, retraining without it. The axis: how much the test RMSE rises, in points.
- **The census blocks:** permuting them costs 1.3 to 3.8 points; retraining without them, 0.3 to 0.4: they stand in for one another.
- **The department** is the only one not fully replaced: 11.4 versus 2.9.
- **The information comes bundled:** how many distinct things do the 37 columns say? Week 7 compresses them.

A13. The same problem, another explanation

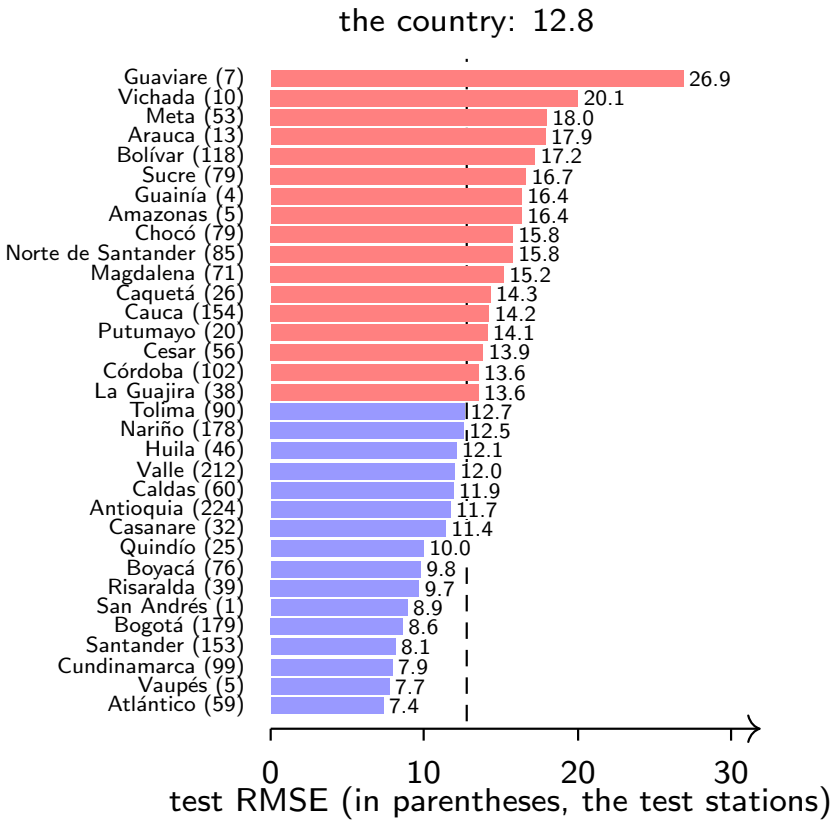
Week 5's boosting with coordinates: almost the same error, another split of the credit



- **With the coordinates,** the error falls from 12.8 to 12.1 and the credit moves out: they take what belonged to the department (from 10.1 to 2.8) and to the Afro share (from 4.0 to 1.5).
- **Which one is the true one?** Neither: two nearly equal models, two stories of the same vote: the Rashomon effect (Breiman, 2001; Fisher et al., 2019).
- **The census, given the map:** without it, 12.6; with it, 12.1.
- **The rule:** an explanation belongs to one model; ask whether another just as good agrees.

A14. The error by subgroup

The importances say what the prediction depends on; this says who it fails



- **By zone:** 9.3 in town centers and 14.9 in rural areas: the error lives in the countryside, where the census says less.
- **By vote level:** where Petro got less than 20, the model predicts 20.1 (actual: 12.5); where he got more than 80, 80.5 (actual: 90.1). The extremes, pulled toward the mean.
- **By department:** worst in Guaviare (26.9, with 7 stations), Vichada, and Meta; best in Atlántico (7.4) and Vaupés.
- **The rule:** without the error by subgroup the report is incomplete; if the error concentrates in one group, the average lies.

A15. Fairness: when the subgroup is a person

- **The per-group metric.** False positives, false negatives, calibration, and precision *by group* (ethnicity, gender, stratum). A model with a high AUC can have twice the false negatives in one group. Our case allocates stations; a model allocating credit or subsidies allocates lives.
- **The label matters more than the algorithm.** Obermeyer et al. (2019): a health model used spending as a proxy for need; Black patients spend less at equal illness and were underestimated. Changing the label raised their share among the prioritized from 17.7 % to 46.5 %.
- **Dropping the sensitive variable is not enough.** Zip code, stratum, and name rebuild it: in our model, dropping the ethnicity block costs 0.3 points, because the department and the rest of the census rebuild it.
- **You cannot have it all.** Per-group calibration and equal error rates are incompatible when prevalence differs (Kleinberg, Mullainathan, and Raghavan, 2017). You must choose, and say which.

For your projects

If the prediction assigns something to people, the per-group metrics table goes next to the test table.

Appendix references and others cited

- Shapley, L. S. (1953). A Value for *n*-Person Games. En *Contributions to the Theory of Games II*, 307–317.
- Strobl, C., Boulesteix, A.-L., Zeileis, A., & Hothorn, T. (2007). Bias in Random Forest Variable Importance Measures. *BMC Bioinformatics*, 8, 25.
- Hooker, G., Mentch, L., & Zhou, S. (2021). Unrestricted Permutation Forces Extrapolation. *Statistics and Computing*, 31, 82.
- Obermeyer, Z., et al. (2019). Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations. *Science*, 366, 447–453.
- Robinson, W. S. (1950). Ecological Correlations and the Behavior of Individuals. *American Sociological Review*, 15(3), 351–357.
- Kleinberg, J., Mullainathan, S., & Raghavan, M. (2017). Inherent Trade-Offs in the Fair Determination of Risk Scores. *ITCS 2017*.
- Strobl, C., Boulesteix, A.-L., Kneib, T., Augustin, T., & Zeileis, A. (2008). Conditional Variable Importance for Random Forests. *BMC Bioinformatics*, 9, 307. [A5]
- Friedman, J., & Popescu, B. (2008). Predictive Learning via Rule Ensembles. *Annals of Applied Statistics*, 2(3), 916–954. [A6]
- Shorrocks, A. F. (2013). Decomposition Procedures for Distributional Analysis: A Unified Framework Based on the Shapley Value. *Journal of Economic Inequality*, 11, 99–126. [A1]
- Chen, H., Janizek, J., Lundberg, S., & Lee, S.-I. (2020). True to the Model or True to the Data? *arXiv*:2006.16234. [A2]
- Joseph, A. (2019). Shapley Regressions: A Framework for Statistical Inference on Machine Learning Models. *Bank of England Staff Working Paper* 784. [A5]